Notes on Bernstein-Sato Polynomials

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1 Introducing the Weyl Algebra A_n

Letting $K[X] = K[x_1, \ldots, x_n]$ denote the ring of polynomials on n-variables. We denote by $End_K(K[X])$ the algebra of linear operators on $K[X]$, that is, linear maps

$$\phi : K[X] \to K[X]$$

equipped with ordinary function addition and multiplication given by concatenation.
We now define the **Weyl Algebra** A_n to be the subalgebra of $End_K(K[X])$ which has generating set

$$\{\hat{x}_1, \ldots, \hat{x}_n, \partial_1, \ldots, \partial_n\} \quad \text{where} \quad \hat{x}_i(p) = x_i \cdot p \quad \text{and} \quad \partial_i(p) = \frac{\partial p}{\partial x_i}$$

It is clear from inspection that for $k \leqslant m$, A_k is a subalgebra of A_n. It can easily be seen that A_n is not commutative with

$$(\partial_i \cdot \hat{x}_i)(f) = \partial_i(x_i \cdot f) = x_i \frac{\partial f}{\partial x_i} + f \qquad \text{whereas} \qquad (\hat{x}_i \cdot \partial_i)(f) = x_i \frac{\partial f}{\partial x_i}$$

This tells us that $\partial_i \cdot \hat{x}_i = (\hat{x}_i \cdot \partial_i) + 1$ which may be expressed in the language of commutators as $[\partial_i \cdot \hat{x}_i] = 1$.

It is straightforward to show that the set

$$\{\hat{x}^\alpha \partial^\beta \mid \alpha, \beta \in \mathbb{N}^n\} \qquad \text{where} \qquad \hat{x}^\alpha \partial^\beta = \hat{x}_1^{\alpha_1} \cdots \hat{x}_n^{\alpha_n} \partial_1^{\beta_1} \cdots \partial_n^{\beta_n}$$

forms a basis for A_n as a K-vector space. We will occasionally deploy the notation $A_n(K)$ to indicate which field our algebra is defined over.

There is a notion of degree for elements of A_n that will be useful later. We define the degree of $P \in A_n$ to be

$$deg(P) := max\left\{ \sum_{i=1}^{n} \alpha_i + \beta_i \;\middle|\; c_{\alpha,\beta} \neq 0 \right\}$$

this can be seen in a few examples:

- $10 + \hat{x}_1^2 \partial_2^4 + \partial_1$ has degree 6
- $\hat{x}_1^2 \hat{x}_2^5 \partial_3 + \hat{x}_1 \partial_1^7 \partial_2$ has degree 10

2 Recalling some relevant algebraic constructions

We quickly recall some aspects of module theory that will be used in the next section. We say that R is a **graded ring** if we can decompose R as a direct sum of additive groups

$$R = \bigoplus_{i \geqslant 0} R_i$$

with the additional property that $R_jR_k \subseteq R_{j+k}$. We say that a module M over a graded ring R is a **graded module** if it may likewise be decomposed as a direct sum of subgroups

$$M = \bigoplus_{i \geqslant 0} M_i$$

and satisfying $R_jM_k \subseteq M_{j+k}$.

Definition. *Let R be a K-algebra. A family $\mathcal{F} = \{F_i\}_{i \geqslant 0}$ of K-vector spaces is a **filtration** of R if*

1. $F_0 \subseteq F_1 \subseteq F_2 \subseteq \cdots \subseteq R$
2. $R = \bigcup_{i \geqslant 0} F_i$
3. $F_i \cdot F_j \subseteq F_{i+j}$

it is easy to show that every graded algebra is filtered.

On the other hand, we can also show that every graded K-algebra has a filtration. Given $G = \bigoplus_{i \geqslant 0} G_i$ is a graded K-algebra, we can see that the vector spaces

$$F_k = \bigoplus_{i=0}^{k} G_i$$

clearly satisfy the increasing nested condition. It is also clear that their union is G. Finally, since

$$F_kF_m = \bigoplus_{i+j \leqslant k+m} G_iG_j$$

and $G_iG_j \subset G_{i+j} \subset \bigoplus_{h=0}^{i+j} G_h = F_{i+j}$, we conclude that $F_kF_m \subseteq F_{k+m}$ and so the F_i give a filtration on G.

Furthermore given a K-algebra R with a filtration $\mathcal{F} = \{F_i\}_{i \geqslant 0}$, we can construct a graded algebra known as the **associate graded algbera** by examining the canonical projections of vector spaces

$$\sigma_k : F_k \to F_k/F_{k-1}$$

and then looking to the K-vector space

$$gr^{\mathcal{F}} R = \bigoplus_{i \geqslant 0} (F_i/F_{i-1})$$

which we equip with a multiplication such that given $a \in F_k$ and $b \in F_m$, we have

$$\sigma_k(a)\sigma_m(b) = \sigma_{m+k}(ab)$$

which is all that is needed to turn this into a graded K-algebra. The construction of a filtration on a K-algebra has analogous ones for rings (each object of the filtration is an ideal) and modules over filtered rings. At risk of being too explicit, we quickly list them.

Definition. *A filtration of a ring R is a sequence of ideals $\mathcal{I} = \{I_i\}_{i \geqslant 0}$ satisfying conditions identical to those of filtered algebras.*

Definition. *Given a filtered ring R and an R-module M, a filtration on M is a sequence of submodules $\Gamma = \{\Gamma_i\}_{i \geqslant 0}$ satisfying*

1. $\Gamma_0 \subseteq \Gamma_1 \subseteq \cdots \subseteq M$
2. $\bigcup_{i \geqslant 0} \Gamma_i = M$
3. $I_n\Gamma_m \subseteq \Gamma_{n+m}$

These both have associate graded objects that are defined analogously to that of the associate graded algebra, noting that the associate graded module $gr^{\Gamma} M$ of a filtered R-module M with filtration Γ is a module over $gr^{\mathcal{I}}(R)$, the associate graded ring of R. Seeing this amounts to familiarizing oneself with the induced action of a quotient ring on modules with an eye towards the third criterion in the definition of a filtered module.

Bernstein's Inequality

An essential tool in the proof of the existence of Bernstein-Sato polynomials is Bernstein's inequality which gives us, in some sense to be made clear, a lower bound on the "dimension" of finitely generated A_n-modules.

Definition. *A **Hilbert polynomial** is a polynomial $\chi(t) \in \mathbb{Q}[t]$ satisfying the condition that there is a positive integer N such that for all $s > N$, we have*

$$\sum_{i=1}^{s} dim(M_i) = \chi(s)$$

That such a thing exists may seem a bit odd (it certainly did to me anyway). Proving that these exist requires some definitions and a technical lemma. For $f : \mathbb{C} \to \mathbb{C}$, we define the **difference function**

$$\Delta f(z) = f(z+1) - f(z)$$

Definition. *We say that $p(t) \in \mathbb{Q}[t]$ is a **numerical polynomial** if there is some positive integer N such that for all $n \geqslant N$, we have $p(n) \in \mathbb{Z}$.*

We also introduce a variation on the binomial coefficient: letting $r \in \mathbb{Z}^{\geqslant 0}$ and t a variable, we define

$$\binom{t}{r} = \frac{t(t-1)\cdots(t-r+1)}{r!}$$

call the above a ***t*-binomial**.[1]

Lemma. *The following hold:*

1. *The numerical polynomials are a free abelian group generated by t-binomials. Furthermore, for a numerical polynomial $p(t)$, we have $p(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$.*
2. *Given some $f : \mathbb{Z} \to \mathbb{Z}$, suppose there is a numerical polynomial q such that $\Delta f(n) = q(n)$ all n past a sufficiently large N. Then there is another numerical polynomial p such that $f(n) = p(n)$.*

Proof. For the first part, we proceed by induction on the degree of some numerical polynomial $p(t)$. If the degree is zero, then $p(t)$ is an integer, so setting $c_0 = p(t)$ gives us the desired equality.

For the inductive step, suppose the result holds for all polynomials of degree less than or equal to $k-1$. If $p(t)$ is a polynomial of degree k, we note that since the leading term of the generalized binomial $\binom{t}{k}$ is $\frac{t^k}{k!}$, it follows that there are $c_0, \ldots, c_k \in \mathbb{Q}$ such that

$$p(t) = \sum_{i=0}^{k} c_{k-i} \binom{t}{i}$$

We now observe that

$$\Delta \binom{t}{i} = \binom{t+1}{i} - \binom{t}{i} = \frac{(t+1)(t)\cdots(t+2-i)}{i!} - \frac{t(t-1)\cdots(t-i+1)}{i!}$$

for brevity, let $g(t) = t(t-1)\cdots(t+2-i)$, we continue

$$\frac{(t+1)g(t) - g(t)(t-i+1)}{i!} = \frac{ig(t)}{i!} = \frac{t(t-1)\cdots(t-i+2)}{(i-1)!} = \frac{t(t-1)\cdots(t-(i-1))}{(i-1)!} = \binom{t}{i-1}$$

with this in hand, we now note that

$$\Delta p(t) = \sum_{i=0}^{k} c_{k-i} \left[\binom{t+1}{i} - \binom{t}{i} \right] = \sum_{i=0}^{k} c_{k-i} \binom{t}{i-1}$$

[1] Apologies if this is confusing, but I made this terminology up because I wanted to be able to refer to it with normal language.

we note that the summand of maximal degree is $c_0\binom{t}{k-1}$, which has degree $k-1$, and so the inductive hypothesis implies that $c_{k-1}, \ldots, c_0 \in \mathbb{Z}$. It follows then that $p(t) - c_k$ is integer valued. Since $p(t)$ a numerical polynomial, it follows that for sufficiently large positve n, we have $p(n)$ an integer as well as $p(n) - c_k$, which means that c_k an integer, proving the condition for numerical polynomials of all degrees.

For the second part of the lemma, we apply the first part to conclude that

$$q(t) = \sum_{i=0}^{k} c_{k-i}\binom{t}{i}$$

with $c_k, \ldots, c_0 \in \mathbb{Z}$. We now let

$$p(t) = \sum_{i=0}^{k} c_{k-i}\binom{t}{i+1}$$

It follows immediately from our computation of $\partial\binom{t}{i}$ in the first part that $\Delta p(t) = q(t)$. It follorws then that for sufficiently large n, we have $\Delta(f-p)(n) = 0$. But this means that

$$(f-p)(n+1) = (f-p)(n)$$

and so $(f-p)(n)$ is a constant integer, which we will call c_{k+1}, for sufficiently large positive n. It follows then that $f(n) = p(n) + c_{k+1}$ for sufficiently large n. ■

with this out of the way, we now deliver on our promise

Theorem. *Given a finitely generated graded module $M = \bigoplus_{i \geqslant 0} M_i$, its Hilbert polynomial exists and is unique.*

Proof. We proceed by induction on the number of variables in our polynomial ring $K[X]$. In the case with zero variables, we are reduced to the base field K. it follows then that out finitely generated module M is a K-vector space and hence only has finitely many homogeneous components, which means that for a sufficiently large integer s, we have

$$\sum_{i=0}^{s} dim_K(M_i) = dim_K(M)$$

, and so the desired polynomial is just the constant polynomial $\chi(t) = dim_K(M)$.

Now suppose that the theorem holds for any finitely generated module over the polynomial ring on $n-1$ variables. Given a finitely generated graded module $M = \bigoplus_{i \geqslant 0} M_i$ over $K[x_1, \ldots, x_n]$, define $f : \mathbb{Z} \to \mathbb{Z}$ by

$$f(s) = \sum_{-\infty}^{i} M_i \qquad \forall k < 0 : M_k = 0$$

letting $\phi_i : M_i \to M_i$ be the linear map defined by multiplication by x_n, set $Q_i = ker(\phi_i)$ and $L_i = coker(\phi_i)$. Consider the exact sequence

$$0 \to Q_{i-1} \to M_{i-1} \xrightarrow{\phi_{i-1}} M_i \to L_i \to 0 \qquad (*)$$

$Q = \bigoplus_{i \geqslant 0} Q_i$ and $L = \bigoplus_{i \geqslant 0} L_i$ are finitely generated graded modules over $K[x_1, \ldots, x_n]$ and are the kernel and cokernel of the endomorphism of M defined by multiplication by x_n. We note that multiplication by x_n annihilates each element of Q and, since multiplication by x_n produces an element of $Im(\phi_{i-1})$, we know that multiplication annihilates L as well. It follows then that Q and L are finitely graded $K[x_1, \ldots, x_n]$-modules.

By the inductive hypothesis, there are Hilbert polynomials $\chi_Q(t), \chi_L(t) \in \mathbb{Q}[t]$ such that for some sufficiently large positive integer s, we have

$$\chi_Q(s) = \sum_{i=0}^{s} dim_K(Q_i) \qquad \chi_L(s) = \sum_{i=0}^{s} dim_K(L_i)$$

we observe also that the dimensions of the vector spaces in $(*)$ are such that

$$dim_K(Q_{i-1}) - dim_K(M_{i-1}) + dim_K(M_i) - dim_K(L_i) = 0$$

Summing these across $0 \leqslant i \leqslant s$ for some sufficiently large s yields

$$\chi_Q(s-1) + \Delta f(s) - \chi_L(s) = 0$$

meaning $\Delta f(s)$ is equal to a difference of numerical polynomials and, hence, is a numerical polynomial. It follows then from part two of the lemma that $f(s)$ is a numerical polynomial. Hence, polynomials satisfying the conditions laid forth in the definition of the Hilbert polynomial exist.

■

We now begin to make use of the properties of filtrations and associate graded algebras.

Definition. *The **Bernstein Filtration** of the nth Weyl algebra is given by the set $\mathcal{B} = \{B_i\}_{i \geqslant 0}$ where we define*

$$B_k = \{P \in A_n \mid deg(P) \leqslant k\}$$

It is clear from inspection that this is a filtration of A_n. Using this, one has an associate graded algebra $\mathfrak{B}_n = gr^B A_n$.

Definition. *We say that a filtration*

$$\mathcal{F} = \{F_i\}_{i \geqslant 0} \qquad \{0\} \subseteq F_0 \subseteq F_1 \subseteq \cdots \subseteq M$$

*of a left A_n-module M is **compatible** with the Bernstein filtration $\mathcal{B} = \{B_i\}_{i \geqslant 0}$ if it satisfies the conditions*

1. $B_j F_k \subseteq F_{j+k}$
2. $M = \bigcup F_i$
3. *Each F_k is finitely generated as a B_0 module.*

We should note that given a finitely generated A_n-module M, the existence of a filtration Γ_i that is compatible with the Bernstein filtration implies for each i, Γ_i is a K vector space (see criterion three of a filtered module and consider left multiplication by $B_0 = K$). In light of this, Coutinho actually imposes a fourth condition for a filtration of a left A_n-module: that each Γ_i is a *finite dimensional* K-vector space.

We are nearly at the stage where we can define what we meant by "dimension" earlier in this section. There is only one more ingredient, that is, noticing that $\mathfrak{B}_n = gr^{\mathcal{B}} A_n$ is isomorphic to the ring of polynomials in $2n$ variables. This isn't difficult to prove, but for brevity we refer the reader to Coutinho. Once we have this, supposing Γ is a filtration of a finitely generated A_n-module M which is compatible with $\mathcal{B}$, we note that $gr^{\Gamma} M$ is a graded module over the (graded) polynomial ring $\mathfrak{B}_n$. That this is the case relies critically on the fact that Γ is compatible with $\mathfrak{B}$.

The above remarks mean that our graded module has a Hilbert polynomial $\chi(t, \Gamma, M)$ which, due to its conditional existence on a compatible filtration, we'll call it a **Bernstein-Hilbert polynomial**.

Definition. *The **dimension** $d(M)$ of a finitely generated A_n-module M is the degree of its associated Bernstein-Hilbert polynomial. Given a_M the leading coefficient of $\chi(t, \Gamma, M)$, we define the **multiplicity** of M to be*

$$m(M) = d(M)! a_M$$

In order for this to be remotely useful, we need it to not depend on our choice of compatible filtration. To show that this notion of dimension satisfies the desired property, we need a lemma.

Lemma. *Letting Γ and Ω be good filtrations of a left A_n-module M, then there is a k such that $\Omega_{j-k} \subseteq \gamma_j \subseteq \Omega_{j+k}$*

This is Coutinho Proposition 8.3.2. The lemma implies that there is a k such that

$$dim_K(\Omega_{j-k}) \leqslant dim_K(\Gamma_j) \leqslant dim_K(\Omega_{j+k})$$

it follows then that for all values t past a sufficiently large postive integer s, we have

$$\chi(t-k, \Omega, M) \leqslant \chi(t, \Gamma, M) \leqslant \chi(t+k, \Omega, M) \qquad (*)$$

We now observe that since the behavior of a polynomial is determined by its term of highest degree, we have

$$\lim_{t\to\infty} \chi(t+k, \Omega, M) - \chi(t-k, \Omega, M) = 0$$

hence the inequality implies that $\chi(t, \Omega, M)$ and $\chi(t, \Gamma, M)$ have the same degree and leading coefficient. Hence, our notion of dimension and multiplicity for finitely generated A_n-modules is independent of our selection of compatible filtration.

Some properties of this that we will make use of include the following:

Given M a finitely generated left A_n-module and N a submodule, we have

1. $d(M) = max\{d(N), d(M/N)\}$
2. If $d(N) = d(M/N)$ then $m(M) = m(N) + m(M/N)$

The proof of this isn't difficult, but is omitted due to time constraints. This is Coutinho Theorem 9.3.2. Using it, we can conclude that given $M_1, \ldots, M_k$ are finitely generated A_n-modules, then if $M = \bigoplus_{i\geqslant 0} M_i$, we have

$$d(M) = max\{d(M_1), \ldots, d(M_k)\}$$

and, furthermore, that if $d(M) = d(M_i)$ for all $1 \leqslant i \leqslant k$, we have

$$m(M) = \sum_{i=1}^{k} m(M_i)$$

Furthermore, the maximum possible dimension of a left A_n module is $2n$. With these remarks in hand, we are finally ready to tackle Bernstein's inequality.

Theorem (Bernstein's Inequality)**.** *If M is a finitely generated non-zero left A_n-module, then $d(M) \geqslant n$.*

Proof. Let $m_1, \ldots, m_k$ be a generating set for M and let Γ be the filtration defined by

$$\Gamma_i = \sum_{j=1}^{k} B_k m_j$$

It is clear that this filtration is compatible and, furthermore, Γ_0 is the k-dimensional vector space generated by the m_i. Let $\chi(t, \Gamma, M)$ be Bernstein-Hilbert polynomial. we would like to show that we have an embedding

$$B_i \to Hom_K(\Gamma_i, \Gamma_{2i})$$

and so we prove a lemma

Lemma. *Given a finitely generated left A_n-module with a compatible filtration Γ with $\Gamma_i \neq 0$, the map*

$$\phi : B_i \to Hom_K(\Gamma_i, \Gamma_{2i})$$
$$a \mapsto \phi_a(u) = au$$

is an embedding.

Proof. This is equivalent to showing that left multiplication of elements of Γ_i by a nonzero $a \in B_i$ is always nonzero. We proceed by induction on i. If $i = 0$, then we're done because $B_0 = K$ and as $\Gamma_0 \neq 0$ by hypothesis, we know that left action by B_0 is non-trivial.

Now suppose the result holds up to $k-1$. Given $a \in B_k$ with a non-zero. Suppose that $a\Gamma_k = 0$. We can immediately conclude that $a \notin K$, hence it follows that a contains as a summand a non-zero monomial $c_{\alpha,\beta}\hat{x}^{\alpha}\partial^{\beta}$. This means that at least one of $\alpha, \beta \in \mathbb{N}^n$ having non-zero indices. If $\alpha_i \neq 0$, then it follows that $[a, \partial_i]$ contains a summand $\alpha_i c_{\alpha,\beta}\hat{x}^{\alpha - e_i}\partial^{\beta}$ which is non-zero. The construction of the commutator guarantees that all terms are either effected by composition with ∂_i or made zero, so we know that $[a, \partial_i] \in B_{k-1}$. Since $a\Gamma_k = 0$ by assumption, we know that

$$[a, \partial_i]\Gamma_{k-1} \subseteq a\partial_i\Gamma_{k-1}$$

but as $\partial_i\Gamma_{k-1} \subseteq \Gamma_k$, we conclude $[a, \partial_i]\Gamma_{k-1} = 0$, contradicting the inductive hypothesis. Hence, the map described is an embedding. ■

It follows from the lemma that

$$dim_K(B_i) \leqslant dim_K(Hom_K(\Gamma_i, \Gamma_{2i}))$$

We now note that since $Hom_K(\Gamma_i, \Gamma_{2i})$ has dimension $dim_K(\Gamma_i) \cdot dim_K(\Gamma_{2i})$ we have

$$dim_K(B_i) \leqslant \chi(i)\chi(2i)$$

for sufficiently large i. We now note that $dim_K(B_i) = \left(\binom{2n+1}{i}\right) = \binom{2n+i}{i} = \binom{2n+i}{2n}$.[2] Viewing this last binomial coefficient as an i-binomial, it generates a polynomial in the variable i of degree $2n$. Thus, as a polynomial in degree i, we know that $\chi(i)\chi(2i)$ must have degree greater than or equal to $2n$. The degree of $\chi(i)\chi(2i)$ is equal to $2d(M)$, we conclude that $d(M) \geqslant n$, as desired. ■

Bernstein-Sato Polynomials

To give a proof of the existence of Bernstein-Sato polynomials, we need to have some discussion of what are called holonomic modules.

Definition. *A finitely generated left A_n-module is called holonomic if it is zero or has dimension n.*

We note that by Bernstein's inequality, holonomic modules are the smallest finitely generated A_n-modules, in some sense. Holonomic modules have some interesting properties that will be useful.

Proposition. *Submodules, quotients, and finite sums of holonomic A_n-modules are holonomic.*

Proof. Given M a holonomic A_n module, let N be a submodule of M. It is clear that $d(N) \leqslant d(M)$ and $d(N/M) \leqslant d(M)$ due to the fact that $d(M) = max\{d(N), d(M/N)\}$. Since $d(M) = n$, we apply Bernstein's inequality to conclude that $d(N) = d(M/N) = n$. The second follows from an application of the first part of this proposition along with the face that given $M \cong \bigoplus_{i=0}^{k} M_i$, we have

$$d(M) = max\{d(M_1), \ldots, d(M_k)\} = n$$

if each M_i is holonomic. ■

Proposition. *Holonomic A_n-modules are Artinian, that is, given a descending sequence of submodules*

$$N_1 \supseteq N_2 \supseteq \cdots$$

this sequence eventually terminates in the sense that there is a k such that for all $j \geqslant k$, we have $N_j = N_k$.

[2] Stars and bars!

Proof. Given M an holonomic A_n-module and a descending sequence of submodules

$$N_0 \supseteq N_1 \supseteq \cdots \supset N_r$$

we know that $= d(N_i/N_{i+1}) = d(N_i) = d(M)$ as M holonomic and since $N_i = N_{i+1} \bigoplus N_i/N_{i+1}$, a result we discussed earlier tells us that

$$m(N_i) = m(N_{i+1}) + m(N_i/N_{i+1}) \qquad (*)$$

and, if the descending sequence never terminates, this would imply that $m(N_i) \geqslant 1$ for all i. This means we can apply one of our earlier discussed properties of summands of to $(*)$ to conclude

$$m(M) = \sum_{i=0}^{r-1} m(N_i/N_{i+1}) + m(N_r) \geqslant r$$

meaning that the number of non-trivial submodules in a descending chain cannot exceed the multiplicity of M. hence, M is Artinian. ■

Given a module M that is Artinian and Noetherian and a sequence of submodules

$$N_0 \subseteq N_1 \subseteq N_2 \subseteq \cdots \subseteq N_k$$

such that N_{i+1}/N_i is irreducible, it follows from a basic property of Artinian modules that there is a submodule N_{k+1} of M such that N_{k+1}/N_k is minimal in the collection of non-zero submodules M/N_k and hence is irreducible. It follows then that the chain

$$N_0 \subseteq \cdots \subseteq N_k \subseteq N_{k+1}$$

satisfies the proeprty that the quotient of adjacent modules is irreducible. Since M is Noetherian, it follows that this process cannot continue indefinitely and hence, we have a sequence

$$0 = N_0 \subseteq \cdots \subseteq N_r = M$$

such that N_{i+1}/N_i is irreducible. This sequence is called a **composition series** for M. It can be shown (Artin-Shreier) that any two composition series for our module have the same number of submodules. This number is called the **length** of M.

Given some $p \in K[X]$, we denote by $K[X, p^{-1}]$ the set of rational functions of the form f/p^r where $f \in K[X]$ is a polynomial. The quotient rule for differentiation tells us that applying a partial derivative to f/p^r will give some polynomial over p^{2r}, which means that $K[X, p^{-1}]$ is a left A_n-submodule of the collection of rational polynomials over K, hereafter denoted $K(X)$.

Theorem. *The A_n-module $K[X, p^{-1}]$ is holonomic.*

Proof. Setting $M = K[X, p^{-1}]$, suppose that p is a polynomial of degree m. Set

$$\Gamma_k = \{f/p^k \mid deg(f) \leqslant (m+1)k\}$$

We'd like to show that these Γ_k form a filtration on M. Given f/p^k an element of M with f having degree s. Expressing this as fp^s/p^{k+s}, we note that fp^s has desgree $s(m+1) \leqslant (m+1(s+k)$. It follws that $f/p^s \in \Gamma_{s+k}$ and that $M = \bigcup_{k \geqslant 0} \Gamma_k$. Given $f/p^k \in \Gamma_k$, we know that $deg(f) \leqslant (m+1)k$. Multiplying f by x_i increases the degree by 1, so $x_i(f/p^k) = x_i fp/p^{k+1 \in \Gamma_{k+1}}$. We also see that

$$\partial_i(f/p^k) = \frac{p\partial_i(f)) - kf\partial_i(p)}{p^{k+1}}$$

the numerator here has degree $\leqslant (m+1)k + (m-1))$, and so we have $\partial_i(f/p^k) \in \Gamma_{k+1}$. This means that

$$B_1\Gamma_k \subseteq \Gamma_{k+1}$$

and since $B_i = \bigoplus_{k=0}^{i} B_1$, we know that $B_1\Gamma_k \subseteq \Gamma_{i+k}$.

Since the dimension of Γ_k cannot be larger than the dimension of the vector space of polynomials of degree $\leqslant (m+1)k$, we conclude that Γ_k is finite dimensional for all k and hence, we know that they form a filtration of M. Furthermore, this shows us that

$$dim_K(\Gamma_k) \leqslant \binom{(m+1)k+n}{n}$$

Examining this as a k-binomial, we see that the resulting polynomial in k that the two terms of highest degree are

$$\frac{(m+1)^n k^n}{n!} \qquad \text{and} \qquad \frac{(m+1)^{n-1}(n+1)nk^{n-1}}{2(n!)}$$

from which we can conclude that for large values of k, we have

$$dim_K(\Gamma_k) \leqslant \frac{(m+1)^n k^n}{n!} + \frac{(m+1)^{n-1}(n+1)n(k+1)^{n-1}}{n!}$$

A rather technical lemma will allow us to conclude the proof

Lemma. *Given M a left A_n-module and a compatible filtration Γ and supposing there are constants c_1, c_2 such that for a large enough positive j, we have*

$$dim_K(\Gamma_j) \leqslant \frac{c_1 j^n}{n!} + c_2(j+1)^{n-1}$$

then M is holonomic.

We omit the proof and refer the reader to Coutinho Lemma 10.3.1. Once persuaded of the lemma, we immediately conclude that M is a holonomic module. ■

We finally come to the purpose of these notes. Bernstein-Sato polynomials are the unique monic polynomials representing one half of a solution to a particular equation, the other half being a differential operator $D(s) \in A_n(K)[s]$, which is just a polynomial in the variable s with coefficients in $A_n(K)$. Specifically, given some $p \in K[X] = K[x_1, \ldots, x_n]$ and a variable s, they solve the equation

$$B(s)p^s = D(s)p^{s+1}$$

To apply the theory we've developed so far to show that such things exist, we need to do just a little more work.

Letting $p \in K[X]$ and s a variable, we begin to construct a holonomic module over $A_n(K(s)) \subset End_{K(s)}(K[X])$. We specify that this module be generated by p^s on which $\hat{x}_i$ acts in the obvious manner and ∂_i acts by

$$\partial_i \cdot p^s = sp^{-1}\frac{\partial p}{\partial x_i} \cdot p^s$$

with multiplication thus defined, our newly constructed cyclic module, $A_n(K(s)) \cdot p^s$, is a submodule of $K(s)[X, p^{-1}] \cdot p^s$ when the latter is viewed as a module over $A_n(K(s))$. The notation here is getting a bit byzantine, so to bring things back down to earth, the elements of $K(s)[X, p^{-1}] \cdot p^s$ are precisely rational functions in the variables $x_1, \ldots, x_n, s$ having form

$$\frac{F_{K(s)}(x_1, \ldots, x_n)}{p^r} \cdot p^s$$

where $F_{K(s)}(x_1, \ldots, x_n)$ a polynomial whose coefficients are rational polynomials in the variable s. We are now ready to prove the theorem.

Theorem (Existence and Uniqueness of Bernstein-Sato polynomials)**.** *Proof.* The first step in the proof is demonstrating that the $A_n(K(s))$-module $K(s)[X, p^{-1}] \cdot p^s$ is holonomic. This ends up following the same argument given in for the theorem above. Since $A_n(K(s)) \cdot p^s$ is a submodule of the holonomic module $K(s)[X, p^{-1}] \cdot p^s$, it must be holonomic and therefore Artinian. It follows that the sequence

$$A_n(K(s))p^s \supseteq A_n(K(s))p \cdot p^s \supseteq A_n(K(s))p^2 \cdot p^s \supset \cdots$$

eventually terminates, meaning there is a $k > 0$ such that

$$p^k p^s \in A_n(K(s))p^{k+1} \cdot p^s$$

We now consider the linear automorphism t of $K(s)[X, p^{-1}] \cdot p^s$ defined by

$$t(s^k \cdot p^s) = (s+1)^k p \cdot p^s$$

we can iteratively apply the inverse of this automorphism k times on $p^k \cdot p^s$ as well as on all of $A_n(K(s))p^{k+1} \cdot p^s$ to get

$$t^{-k}(p^k \cdot p^s) = p^k p^{-k} \cdot p^s = p^s \in A_n(K(s))p \cdot p^s$$

which means there is an element of $A_n(K(s))$

$$E = \sum_{(\alpha,\beta)\in\mathbb{N}^n\times\mathbb{N}^n} \frac{P_{(\alpha,\beta)}(s)}{Q_{(\alpha,\beta)}(s)} \hat{x}^\alpha \partial^\beta$$

which satisfies $p^s = Ep \cdot p^s$. Since there are only finitely many non-zero summands comprising E, we can uniformize them with respect to the denominator, an element of $K[s]$. Multiplying both sides by this denominator yields our desired $B(s)$ on the left and $D(s)$ on the right.

The solution pairs $(B(s), D(s))$ are not unique since $B(s) = D(s) = 0$ is a trivial solution.[3] Furthermore, if $B_1(s)$ and $B_2(s)$ with differential operators $D_1(s)$ and $D_2(s)$ satisfy the equation, then their sums will as well. Finally $B(s)$ a solution implies $-B(s)$ also a solution and the product of any solution pair $(B(s), D(s))$ with an element of $P \in K[s]$ will yield a solution pair $(PB(s), PD(s))$. Hence, the polynomials $B(s)$ satisfying $B(s) \cdot p^s \in A_n(K)[s]p \cdot p^s$ form an ideal in the PID $K[s]$, from which it follows that there is a unique monic generating all non-trivial solutions. Therefore, Bernstein-Sato polynomials exist and are unique.

■

[3] We note however that the proof we gave above does not yield a zero solution by virtue of t being an automorphism.